Literature audit reveals one critical v8.3 flaw and two high-value additions

All 10 papers are real, but the single most important finding is a paper the survey didn't cite. DiffusionDriveV2 (arXiv 2512.07745, December 2025) GitHub demonstrates that applying vanilla RL fine-tuning — including DPPO — to an anchor-based diffusion policy causes **mode collapse** unless advantage estimation is performed within each anchor group. Emergent Mind This directly threatens the user's Phase 2 Route A (DPPO on anchor-based policy from Phase 1a). A v8.4 revision is warranted, but the scope is narrow: modify the Phase 2 RL strategy to be anchor-aware, add S²Diff-style Lyapunov certification to Phase 3, and acknowledge the safety-guarantee gap in ManiCM distillation.

Part 1: Verification table for all 10 papers

#	Paper	Real?	Venue / arXiv	Code	v8.3 Relevance
1	iDP3	✅ Real	IROS 2025 / 2410.10803	✅ GitHub	**High** — base architecture
2	CGD	✅ Real	arXiv 2405.01758 (MIT)	❌ None	Medium — constraint-guided diffusion for UAVs, but no code
3	MPD	✅ Real	arXiv 2308.01557, extended Dec 2024	✅ GitHub	Medium — motion planning baseline
4	Aligning Diffusion	✅ Real	arXiv 2504.00342	❌ None	Low — theoretical, very recent, no code
5	OOD-GMixup	✅ Real	IEEE TKDE 2024 / 2308.08344	❌ None	**None** — graph-domain method, not transferable
6	TransDiffuser	✅ Real	arXiv 2505.09315 (Li Auto)	❌ None	Low — decorrelation would *worsen* std ratio issue
7	DAC	✅ Real	ICLR 2025 / 2405.20555	⚠️ JAX only, GitHub (26 stars)	**None** — offline RL, incompatible with online Gazebo setup

#	Paper	Real?	Venue / arXiv	Code	v8.3 Relevance
8	MDPI Drones safety	✅ Real	Drones 8(9), 429 (2024) / DOI:10.3390/drones8090429	❌ None	Low — fixed-wing CPO, not diffusion-compatible
9	DiffusionDrive	✅ Real	CVPR 2025 Highlight / 2411.15139	✅ GitHub	**High** — anchor mechanism basis; **V2 exists**
10	BESO	✅ Real	RSS 2023 / 2304.02532	✅ GitHub	**High** — CFG basis; known issues at high guidance scales

Every paper the AI survey cited is genuine — no hallucinations. However, several papers (OOD-GMixup, TransDiffuser, DAC) were cited misleadingly, implying relevance they don't actually have. The most actionable discoveries came from follow-up works the survey missed entirely.

Part 2: Deep dive on the three highest-value items

DAC is the wrong tool — DPPO wins on every axis

DAC (Diffusion Actor-Critic, ICLR 2025) formulates KL-constrained policy iteration as diffusion noise regression (OpenReview) with "soft Q-guidance" baked into the noise prediction target. (arXiv) It's an elegant offline RL method — but **every architectural decision in DAC opposes the user's setup**.

The fundamental mismatch is paradigmatic: DAC requires a pre-collected offline dataset and constrains the policy to stay near the behavior distribution. (arXiv) The user has a Gazebo Gym wrapper providing online interaction, which is exactly what DPPO was designed for. DAC's conservatism would actively prevent learning recovery behaviors for the overshoot failure mode, because those states are underrepresented in any offline dataset. DPPO's on-policy training means the policy is always learning from its current distribution, naturally addressing covariate shift. (OpenReview) (arXiv)

The technical incompatibilities compound. **DAC cannot fine-tune only the last K denoising steps** — it trains a single noise predictor across all diffusion timesteps simultaneously with shared parameters updated jointly. The user's freeze strategy (AnchorHead + PC encoder + first T-3 steps frozen) is a documented DPPO best practice with a dedicated `ft_denoising_steps` parameter; (arXiv) DAC has no equivalent. (ResearchGate) Additionally, DAC's codebase is JAX-only with 26 GitHub stars and 15 commits, (github) while DPPO is PyTorch-based (GitHub) with **719 stars** and extensive multi-environment configs. (Hugging Face) Porting DAC to PyTorch and adapting it for 3D point cloud observations would take 4–8 weeks with uncertain outcomes versus 1–2 weeks for DPPO integration.

Verdict: Do not adopt DAC. The user's Phase 2 DPPO decision is correct — but must be modified to be anchor-aware (see v8.4 recommendation below).

The MDPI Drones paper is real but architecturally irrelevant

Zhao et al. (2024) is verified at the exact claimed location — MDPI Drones 8(9), 429. (MDPI) It does use Lyapunov stability requirements and rudder deflection limits, exactly as claimed. The method augments Constrained Policy Optimization (CPO) with a Lyapunov-derived safety budget and adds condition-triggered meta-learning for online adaptation. (MDPI)

However, the paper targets **fixed-wing aircraft** with rudder/aileron/elevator dynamics, not quadrotor position/yaw control. It uses traditional CPO — a policy gradient method fundamentally different from diffusion policies. The safety constraints are soft/cumulative (expectation-based), whereas the user needs **hard per-step constraints** ($|\Delta\text{yaw}| \leqslant 30°$). The Lyapunov certification concept is valuable, but the specific implementation cannot transfer.

The far more relevant discovery is **S²Diff** (Cheng et al., NeurIPS 2025, arXiv 2509.25375), which directly combines Lyapunov stability with diffusion policies using Control Lyapunov-Barrier Functions (CLBFs). S²Diff achieves (arXiv) (OpenReview) **98.75% safety rate** across tasks including a 16-dimensional F-16 aircraft model without requiring control-affine dynamics assumptions. (alphaXiv) It establishes an intrinsic connection between diffusion sampling and Almost Lyapunov theory (arXiv) — even if the Lie derivative condition is locally violated, global stability holds if violations are confined to small-volume regions. (arXiv) This is the paper the user actually needs for Phase 3 sim-to-real safety certification.

TransDiffuser would actively harm the user's system

TransDiffuser (arXiv 2505.09315, Li Auto) is real (ResearchGate) and proposes a training-time decorrelation loss (arXiv) inspired by Barlow Twins that penalizes off-diagonal elements of the multi-modal feature correlation matrix. (The Moonlight) It operates purely as a loss regularization term — it does not modify inference. (themoonlight) (The Moonlight)

The critical problem: **TransDiffuser's decorrelation mechanism increases trajectory diversity**, which would *worsen* the user's std ratio problem (prediction std / GT std = 2.24). The user's model already generates overly diverse predictions — a classic **mode-covering** signature where the diffusion model spans multiple behavioral modes simultaneously. TransDiffuser is designed to push features further apart, which would amplify this exact failure mode. Furthermore, TransDiffuser explicitly argues against anchor-based approaches, (arXiv) (ResearchGate) making it philosophically contradictory to the user's Phase 1a architecture. The decorrelation concept itself is well-established (Barlow Twins, VICReg) and doesn't require this specific paper.

DiffusionDriveV2 changes the Phase 2 calculus entirely

The survey's most consequential omission is **DiffusionDriveV2** (arXiv 2512.07745, December 2025), which extends DiffusionDrive with RL fine-tuning and directly addresses the anchor + RL interaction the user faces in Phases 1a → 2.

DiffusionDriveV2's core finding: naively applying vanilla RL (including PPO variants) across all trajectory samples from an anchor-based diffusion policy causes **mode collapse** because advantage estimation compares trajectories with fundamentally different driving intentions (e.g., turning left vs. going straight). Emergent Mind The solution is **Intra-Anchor GRPO** — Group Relative Policy Optimization performed exclusively within each anchor group, plus Inter-Anchor Truncated GRPO for global stability. Emergent Mind arXiv This achieves **91.2 PDMS on NAVSIM v1** (new SOTA) without destroying multimodality. Emergent Mind arXiv

The implications for v8.3 are direct and serious. The user's Phase 2 applies DPPO to a policy built with anchors in Phase 1a Week 1. Without anchor-aware advantage estimation, DPPO would collapse the anchor-structured distribution — exactly the failure DiffusionDriveV2 documents. The user's current "freeze first T-3 steps" strategy mitigates this somewhat by limiting the RL update scope, but does not address the fundamental issue of cross-anchor advantage comparison.

Code and weights for DiffusionDriveV2 are available at github.com/hustvl/DiffusionDriveV2. GitHub

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The user's 3-layer architecture is novel and unvalidated

No published work combines all three of: anchor-based truncated diffusion + Classifier-Free Guidance + constraint projection. Several papers combine two of the three — DiffusionDriveV2 (anchors + RL constraints), SafeDiffuser (diffusion + CBF projection), ICLR OpenReview CoBL-Diffusion (CLF/CBF + diffusion guidance) — but the specific three-layer stack the user proposes in Phase 1a is **architecturally novel**.

This is both opportunity and risk. The most concrete interaction concern is between anchors and CFG: anchors promote **between-mode diversity** by partitioning the action space, arXiv while CFG at high guidance scales **reduces within-mode diversity** and can cause off-manifold sampling. arXiv arXiv Recent work (CFG++, arXiv 2406.08070, ICLR 2025) shows that standard CFG omits a Rényi divergence correction term, leading to over-concentrated samples. Emergent Mind Emergent Mind The user should keep CFG guidance scales moderate and monitor within-anchor diversity metrics during Phase 1a Week 2 integration.

For the projection barrier layer, **Constrained Diffusers** (Zhang et al., NeurIPS 2025, arXiv 2506.12544) provides a directly applicable plug-and-play framework that enforces constraints on pre-trained diffusion models without retraining arXiv — implementing projected, primal-dual, and augmented Lagrangian approaches. arXiv arXiv This could replace or augment the user's custom projection implementation with a more principled and tested framework.

The std ratio of 2.24 is mode covering, not mode collapse

The user's observed prediction std / GT std ratio of 2.24 is consistent with the well-documented

mode-covering property of diffusion models, not mode collapse. Diffusion-QL (Wang et al., ICLR 2023) explicitly describes this: (arXiv) single-mode diffusion policies place "high density in the middle area of two modes, which is actually the low-density region of the training data." (GitHub) A ratio of 2.24 likely indicates the model spans **2–4 distinct behavioral modes** simultaneously rather than selecting one.

No paper specifically identifies 2.24 as a diagnostic threshold. However, the fixes are well-characterized: anchors (DiffusionDrive), (ResearchGate) (TheCVF) CFG at moderate scales (BESO), (Robotics: Science and Syst…) (Intuitive-robots) and anchor-aware RL (DiffusionDriveV2). (arXiv) (arXiv) The user's Phase 1a architecture already deploys the first two, which should reduce the ratio. The key diagnostic tools available are mode diversity scores (as used in DiffusionDrive), per-anchor variance tracking, and clustering analysis of generated trajectories. (Emergent Mind)

Part 3: v8.4 revision recommendation

Yes, produce v8.4. Three specific modifications are warranted:

Modification 1 — Phase 2 Route A: Anchor-aware DPPO (HIGH priority). Replace vanilla DPPO with an anchor-aware variant inspired by DiffusionDriveV2's Intra-Anchor GRPO. Concretely: when computing PPO advantages for the last 3 denoising steps, partition trajectory samples by their nearest anchor and compute advantages within each anchor group. This prevents cross-anchor advantage comparisons that would destroy multimodality. (arXiv) Engineering effort is **medium** — it requires modifying the advantage computation in the DPPO training loop, not the entire architecture. The user should study DiffusionDriveV2's open-source implementation for the exact intra-group normalization scheme.

Modification 2 — Phase 3: Add S²Diff Lyapunov certification layer (MEDIUM priority). After ManiCM consistency distillation, add a CLBF safety filter that certifies trajectory-level Lyapunov stability for the yaw/altitude constraints. This addresses the documented gap that ManiCM provides **no safety guarantees** — its authors explicitly recommend external safety regions. (arXiv) The CLBF would provide a domain-independent stability certificate that transfers across sim and real. (arXiv) (NeurIPS) Engineering effort is **medium-high** (2–3 weeks): define CLBF structure for yaw/altitude, train on simulation data, integrate as post-distillation safety check.

Modification 3 — Phase 1a Week 3: Reference Constrained Diffusers as alternative to custom projection (LOW priority). The user's custom projection barrier is valid — it's validated by Römer et al. (L4DC 2025, "Safe Diffusion Policies with Model-Based Projections"). But Constrained Diffusers (NeurIPS 2025) provides a tested, plug-and-play framework (arXiv) that may save implementation effort. (arXiv +2) This is a convenience improvement, not a correctness fix.

No other sections of v8.3 require modification. The Phase 0 augmentation strategy, Phase 1b Gazebo wrapper, Phase 2 decision matrix thresholds, and ManiCM distillation step are all sound.

Genuinely valuable — implement or study these:

- **DiffusionDriveV2** (not in original 10, discovered during audit): The single highest-value finding. Directly prevents a mode-collapse failure the user would hit in Phase 2. Study the intra-anchor GRPO mechanism and adapt it for DPPO. Open-source code available.
- **S²Diff** (not in original 10, discovered during audit): Provides the Lyapunov-for-diffusion theory the user needs for Phase 3 safety certification. The Almost Lyapunov framework is elegant and practical. (arXiv)
- **DPPO** (Paper 7's comparator, arXiv 2409.00588): Confirmed as the correct choice for Phase 2. (OpenReview) ICLR 2025, (Iclr) (ResearchGate) 719 GitHub stars, PyTorch, supports selective denoising step fine-tuning. (arXiv) (arXiv) No change needed except making it anchor-aware.
- **Constrained Diffusers** (NeurIPS 2025, arXiv 2506.12544): Plug-and-play constraint enforcement that could simplify Phase 1a Week 3 implementation. (arXiv +3)
- **SafeDiffuser** (ICLR 2025, arXiv 2306.00148): CBF-based safety for diffusion with formal finite-time invariance guarantees. (Iclr) (OpenReview) Relevant backup for Phase 1a Week 3 if custom projection proves insufficient.

Nice-to-know but not actionable for v8.3/8.4:

- **CGD** (Paper 2): Directly relevant topic (constrained diffusion for UAVs) but no code. (Papers with Code +2) Follow-up papers note its constraint enforcement "falls short" compared to projection methods. (arXiv) Worth citing but not implementing.
- **BESO** (Paper 10): Already correctly used. Be aware of CFG++ improvements for avoiding off-manifold sampling at high guidance scales. (Intuitive-robots +2) No architectural change needed — just tune guidance scale conservatively.
- **MDPI Drones safety paper** (Paper 8): Real but wrong domain (fixed-wing CPO). (MDPI) The Lyapunov idea is sound; S²Diff executes it better for diffusion policies.
- **MPD** (Paper 3): Useful baseline reference, extended version adds B-spline parameterization. (arXiv) (arXiv) Not actionable for the user's architecture.
- **Aligning Diffusion** (Paper 4): Interesting constraint-alignment theory but too early (no code, limited validation). (arXiv)

Distractions that would waste engineering effort:

- **DAC** (Paper 7): Offline RL method (OpenReview) that is incompatible with the user's online Gazebo setup on every axis — paradigm (offline vs. online), framework (JAX vs. PyTorch), architecture (no selective step fine-tuning), maturity (26 vs. 719 GitHub stars). (arXiv) (arXiv) Adopting DAC would cost 4–8 weeks and produce worse results than DPPO. **Clear reject.**

- **OOD-GMixup** (Paper 5): Graph-domain augmentation method (arXiv) with zero transferability to 3D point clouds without converting the entire pipeline to GNNs. The user's existing Phase 0 augmentations (dropout + jitter + rotation + depth noise) are better matched to the domain. **Clear reject.**
- **TransDiffuser** (Paper 6): Would actively worsen the user's mode-covering problem (std ratio 2.24) by increasing output diversity. Designed to replace anchors, not complement them. (arXiv) The underlying decorrelation concept (Barlow Twins) is available without this paper. (arXiv) **Clear reject.**

The survey's citation quality follows a pattern: the 4 already-verified papers (1–4) are genuinely relevant, the 6 "new" papers include 3 that are irrelevant distractions (5, 6, 7) and 3 that are real but low-impact for this specific architecture (8, 9, 10). The highest-value discoveries — DiffusionDriveV2, S²Diff, Constrained Diffusers, SafeDiffuser — were all **absent from the survey**, underscoring the importance of this independent literature audit before committing to implementation.